

1. Introduction

With the increasing use of smartphones, mobile applications have become essential tools for managing daily activities, including health and fitness. However, many existing mobile fitness applications either focus only on workout tracking or provide generalized diet plans without considering real-world constraints such as user budgets and local food availability.

Similarly, gym management systems are often web-based and lack mobile accessibility, limiting user engagement and convenience. There is a growing need for a mobile-based solution that integrates gym management features with intelligent and practical fitness guidance.

2. Problem Description

The global fitness technology landscape largely follows a "one size fits all" approach, where applications provide generic caloric calculations without considering real-world economic and environmental factors. A major issue arises from the disconnect between theoretically generated diet plans and the practical ability of users to obtain those foods within their local context.

Most applications rely on static mappings between user biometric data (such as TDEE) and predefined meal databases. These systems fail to consider local food availability, fluctuating prices, and individual budget constraints. As a result, users are often presented with diet plans that are unrealistic, expensive, or difficult to maintain.

Additionally, AI-based systems that depend purely on text generation may produce inaccurate or inconsistent outputs when handling strict numerical constraints such as budget limits and nutritional targets. This further reduces the reliability of such systems.

This gap between theoretical fitness planning and real-world feasibility leads to poor user adherence, reduced motivation, and ultimately failure to achieve fitness goals.

3. Motivation

The motivation for this project is to bridge the gap between intelligent fitness planning and real-world practicality. There is a clear need for a system that not only provides accurate and personalized diet recommendations but also ensures that these recommendations are achievable within the user's financial and local constraints.

Furthermore, integrating these capabilities into a mobile-based gym management platform enables users to access workout tracking, administrative services, and personalized diet planning in a single, convenient solution. This improves usability, engagement, and the overall effectiveness of the fitness experience.

The motivation for this project is to create a mobile application that bridges the gap between fitness planning and real-world practicality.

Users need a system that:

- Provides personalized workout and diet plans
- Considers their financial limitations
- Is easily accessible through a mobile device
- Supports continuous tracking and engagement

By integrating intelligent features into a mobile platform, the system aims to improve user adherence to fitness routines and enhance overall user experience.

4. Proposed Solution

The proposed solution is a mobile-based Gym Management System integrated with an AI-powered Diet Planning module.

Key features of the system include:

- Mobile-friendly user interface for easy access
- Member registration and profile management
- Trainer management and class scheduling
- Payment and subscription handling
- Workout tracking and progress monitoring
- Personalized diet plan generation based on user data
- Consideration of food prices and user budget in diet planning

The system uses intelligent processing to generate diet plans that are both nutritionally appropriate and economically feasible, ensuring that users can realistically follow the recommendations.

5. Objectives

Main Objective

To design and develop a mobile application that integrates gym management functionalities with intelligent, budget-aware diet planning.

Specific Objectives

- Develop a user-friendly mobile application for gym members
- Generate personalized diet plans based on user attributes and goals
- Incorporate budget considerations into diet recommendations
- Enable tracking of workouts and user progress

- Provide efficient management of trainers, classes, and equipment
- Improve user engagement through a centralized mobile platform

6. Expected Outcome

The system will provide a complete mobile solution that allows users to manage their fitness journey effectively. Users will receive practical diet plans tailored to their needs and budget, while also being able to track workouts and interact with gym services through a single application.

7. Conclusion

This project addresses the limitations of existing mobile fitness applications by integrating gym management with intelligent and realistic diet planning. The proposed solution ensures a more practical, accessible, and user-centered approach to achieving fitness goals.